The Constrained Jet of the Lorentzian Eliminated Block and the Chiral Generator: From the Schur Reduction to the Transported Equivariance Defect

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Abstract

This note isolates a single operator-level lemma in the fermionic-matter sub-programme of Cosmochrony, logically between the Schur reduction of the projective residue [5] and the physical identification of the chiral generator. The Schur reduction writes the projective endomorphism as $E_\Pi = -\Pi_S\, D\,(1-P)\, D\, \Pi_S^*$, exhibiting the eliminated block $1-P$ as the controlling object of the three-generation split coefficient u; the companion genus analysis fixes the electric sign $\mu_\chi^2 < 0$ [4], while the absolute value $|u|$, the Yukawa sector, and the mass spectrum are kept downstream [1]. We freeze the structural form that the Lorentzian eliminated block must take. Writing the self-adjoint projector family as a jet $1 - P(s) = Q_0 + sQ_1 + s^2Q_2 + O(s^3)$ along the J_Π-odd modulus s, the projector constraints force Q_1 to be a purely off-diagonal Grassmann tangent, $Q_1 = [A, Q_0]$ with $Q_0Q_1Q_0 = 0 = (1-Q_0)Q_1(1-Q_0)$, and pin the diagonal blocks of Q_2 to Q_1^2 with opposite signs. Decomposing the generator by J_Π-parity, $A = A_+ + A_-$, only the chiral part contributes to the split: $\Pi_{J_\Pi\text{-odd}}Q_1 = [A_-, Q_0]$. Propagation through the J_Π-even pair (Π_S, D) grades the residue jet as E_0 even, E_1 odd, E_2 even, so the J_Π-odd part of E_Π^2 at first order is the anticommutator $\{E_0, E_1\}$, and the split source is $u(s) = s\,\langle J_3, \{E_0, E_1\}\rangle / \langle J_3, J_3\rangle + O(s^3)$. With phase-free jet data the mixing coefficient v vanishes identically while u is generically non-zero, so $u \neq 0$ is structurally admissible with no R_{mix} leakage and with $|u|$ left entirely free. We then identify the split-sourcing component of the generator. Because the antiunitary Born–Infeld parity exchanges chiralities, the J_Π-odd part of the eliminated block has chiral-diagonal component equal to the transported equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau\,\overline{\pi_{RR}}\,\tau^{-1}$ of [5], so $[\Pi_{J_\Pi\text{-odd}}(1-P)]_{LL} = \frac{1}{2}\Delta_\chi(P)$ and, along the modulus, $[\Pi_{J_\Pi\text{-odd}}\dot{Q}(0)]_{LL} = \frac{1}{2}\partial_s\Delta_\chi(0)$. The chiral generator A_- is therefore not a new selector but exactly the infinitesimal transported chiral defect rate; its absolute value is not normalised here. All identities are verified by exact symbolic and exact-rational computation, leaving a single downstream target — the admissible normalisation of $\partial_s\Delta_\chi(P)|_0$ — with no circularity against $|u|$, the chiral-frontier normalisation, or the Yukawa sector.

Keywords: eliminated block; Schur complement; self-adjoint projector jet; Grassmann tangent; parity grading; chiral generator; transported equivariance defect; chiral block reduction; generation split; metaplectic phase; non-injective projection.

Contents

1 Position and purpose

The fermionic-matter sub-programme locates the three-generation mass split in the J_3-odd part of the squared projective endomorphism E_Π^2 [3] restricted to the gauge-singlet generation triplet $\mathbb{C}^3_{\rm gen}$, parametrised by a single real number u through $E_\Pi^2|_{\mathbb{C}^3_{\rm gen}} = \mathrm{diag}(1, \frac{1}{2}+u, \frac{1}{2}-u)$, the even sector $\mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$ being closed by Born–Infeld parity [2]. Two results bracket the present note. The Schur reduction [5] exhibits the projective residue as the Schur complement

$$E_\Pi = -\Pi_S\, D\,(1-P)\, D\,\Pi_S^* = -M^\dagger M, \qquad P = \Pi_S^*\Pi_S, \qquad M = (1-P)\, D\,\Pi_S^*, \tag{1}$$

so that the eliminated block $1-P$ is the object controlling u, and closes the Schur transversality in the present Lorentzian spin stratum, giving $u \neq 0$. The companion genus analysis [4] fixes the electric sign $\mu_\chi^2 < 0$ of the saturation curvature, so the split opens spontaneously. What neither result does is fix the explicit operator form of the Lorentzian block $1 - P(s)$, the first open deliverable listed in [5]; the absolute value $|u|$, the Yukawa sector, and the mass spectrum remain downstream and dictionary-bound [1].

This note settles a logically prior and self-contained question: *what constrained form must the eliminated block take, before any generator is identified, for $u \neq 0$ to be possible without importing $|u|$?* The answer is an operator jet lemma. Section 2 sets the jet and its grading. Section 3 proves the constrained-jet theorem: the first-order coefficient is a purely off-diagonal Grassmann tangent, and its only split-sourcing part is the chiral commutator $[A_-, Q_0]$. Section 4 propagates the grading to the residue. Section 5 reads off the split source and its phase-free vanishing of mixing. Section 6 identifies the chiral generator: its J_Π-odd component is the transported chiral equivariance defect of the eliminated block. Section 7 states the scope: $u \neq 0$ is admissible with $|u|$ free, and the only remaining target is the admissible normalisation of the defect rate. Appendix A records the exact verification.

2 The eliminated-block jet and its grading

For each value of the J_Π-odd modulus s opened by Lorentzian complexification, the spinorial embedding $\Pi_S(s)$ defines an orthogonal projector $P(s) = \Pi_S(s)^*\Pi_S(s)$ onto the retained spinorial directions, so the eliminated block $1 - P(s)$ is a self-adjoint projector:

$$(1 - P(s))^2 = 1 - P(s), \qquad (1 - P(s))^\dagger = 1 - P(s). \tag{2}$$

We study its jet at the finite-locking point $s = 0$,

$$1 - P(s) = Q_0 + sQ_1 + s^2 Q_2 + O(s^3), \qquad Q_k = Q_k^\dagger. \tag{3}$$

A smooth self-adjoint-projector family is locally a unitary conjugation orbit of its base point: there is an anti-self-adjoint generator $A = -A^\dagger$ with

$$1 - P(s) = U(s)\, Q_0\, U(s)^\dagger, \qquad U(s) = \exp(sA), \tag{4}$$

whose Taylor coefficients are $Q_1 = [A, Q_0]$ and $Q_2 = \frac{1}{2}[A, [A, Q_0]]$. The chiral involution J_Π (with $J_\Pi^2 = 1$) grades operators into even and odd parts; the finite block is even.

Definition 1 (J_Π-parity)**.** An operator X is J_Π-even if $J_\Pi X J_\Pi = X$ and J_Π-odd if $J_\Pi X J_\Pi = -X$. The odd projection is $\Pi_{J_\Pi\text{-odd}} X = \frac{1}{2}(X - J_\Pi X J_\Pi)$. The base point is even, $J_\Pi Q_0 J_\Pi = Q_0$, and the generator splits as $A = A_+ + A_-$ with $J_\Pi A_\pm J_\Pi = \pm A_\pm$.

The base point Q_0 is the finite, symmetric eliminated subspace. Because it is J_Π-even, E_0^2 has no J_3-odd part and $u(0) = 0$: the split cannot be a finite-fibre datum, recovering the finite-locking equivariance $u_{\text{fin}} = 0$ of [5] and the statement that chirality is a Lorentzian, not a finite, label.

3 The constrained jet

Theorem 1 (Constrained jet of the eliminated block)**.** *Let $1 - P(s)$ be a smooth self-adjoint-projector family* (2) *with jet* (3) *and generator A as in* (4)*. Then:*

(i) The first-order coefficient is purely off-diagonal with respect to Q_0,

$$Q_0 Q_1 Q_0 = 0 = (1-Q_0) Q_1 (1-Q_0), \qquad Q_1 = Q_0 Q_1 (1-Q_0) + (1-Q_0) Q_1 Q_0 = [A, Q_0]. \tag{5}$$

(ii) Decomposing the generator by J_Π-parity, $Q_1 = [A_+, Q_0] + [A_-, Q_0]$ with $[A_+, Q_0]$ J_Π-even and $[A_-, Q_0]$ J_Π-odd, so the split-sourcing part is

$$\Pi_{J_\Pi\text{-}odd} Q_1 = [A_-, Q_0]. \tag{6}$$

(iii) The diagonal blocks of the second-order coefficient are pinned by Q_1^2 with opposite signs,

$$Q_0 Q_2 Q_0 = -Q_0 Q_1^2 Q_0, \qquad (1-Q_0) Q_2 (1-Q_0) = +(1-Q_0) Q_1^2 (1-Q_0), \tag{7}$$

with $Q_2 = \frac{1}{2}[A,[A,Q_0]]$ J_Π-even; only the off-diagonal even part of Q_2 is free.

Proof. Expand $(1-P(s))^2 = 1-P(s)$ order by order. Order s^1 gives $Q_0Q_1 + Q_1Q_0 = Q_1$. Sandwiching by Q_0 yields $2Q_0Q_1Q_0 = Q_0Q_1Q_0$, hence $Q_0Q_1Q_0 = 0$; sandwiching by $1-Q_0$ yields $(1-Q_0)Q_1(1-Q_0) = 0$. The two off-diagonal blocks then exhaust Q_1, which is (5); the conjugation form gives $Q_1 = [A, Q_0]$ directly. For (ii), $J_\Pi[A_\pm, Q_0]J_\Pi = [J_\Pi A_\pm J_\Pi, Q_0] = \pm[A_\pm, Q_0]$ since Q_0 is J_Π-even, and $\Pi_{J_\Pi\text{-odd}}$ extracts the odd summand $[A_-, Q_0]$. Order s^2 gives $Q_0Q_2 + Q_2Q_0 + Q_1^2 = Q_2$. Sandwiching by Q_0 yields $2Q_0Q_2Q_0 + Q_0Q_1^2Q_0 = Q_0Q_2Q_0$, hence $Q_0Q_2Q_0 = -Q_0Q_1^2Q_0$; sandwiching by $1-Q_0$ leaves only $(1-Q_0)Q_1^2(1-Q_0)$ on the left, equal to $(1-Q_0)Q_2(1-Q_0)$, which is the second identity of (7). The conjugation form gives $Q_2 = \frac{1}{2}[A,[A,Q_0]]$, J_Π-even by Definition 1. □

Remark 1 (The split-killing condition is on A_-, not on Q_1). A purely J_Π-even generator leaves $Q_1 = [A_+, Q_0]$ generically non-zero while contributing nothing to the split, since $\Pi_{J_\Pi\text{-odd}}Q_1 = 0$. The split is therefore dead if and only if $[A_-, Q_0] = 0$ — for instance if A_- carries no off-diagonal component between Q_0 and $1-Q_0$ — and not merely if $Q_1 = 0$. The single structural degree of freedom that can open the generation split is the chiral Grassmann tangent $[A_-, Q_0]$.

4 Propagation to the residue

Lemma 1 (Parity propagation)**.** *If Π_S and D commute with J_Π, then the residue jet $E_\Pi(s) = E_0 + sE_1 + s^2E_2 + O(s^3)$ induced by $E_k = -\Pi_S\, D\, Q_k\, D\, \Pi_S^*$ from* (1) *satisfies $J_\Pi E_k J_\Pi = (J_\Pi Q_k J_\Pi\text{-parity})\, E_k$; explicitly E_0 is J_Π-even, E_1 is J_Π-odd, E_2 is J_Π-even.*

Proof. Conjugation by a J_Π-even operator preserves J_Π-parity: if $J_\Pi G J_\Pi = G$ and $J_\Pi X J_\Pi = \varepsilon X$, then $J_\Pi(GXG^\dagger)J_\Pi = (J_\Pi G J_\Pi)(J_\Pi X J_\Pi)(J_\Pi G^\dagger J_\Pi) = \varepsilon\, GXG^\dagger$. Apply this with $G = \Pi_S D$ and $X = Q_k$ of parity $(+,-,+)$ for $k = 0,1,2$ from Theorem 1. □

5 The split source

Restrict to the generation triplet $\mathbb{C}^3_{\mathrm{gen}}$ in the frozen basis (e_0, e_+, e_-) with $J_3 = \mathrm{diag}(0,1,-1)$, R_{mix} the off-diagonal antisymmetric generator on (e_+, e_-), and $J_\Pi^{(2)} : e_0 \mapsto -e_0,\ e_+ \leftrightarrow e_-$. The Hilbert–Schmidt pairing is $\langle X, Y\rangle = \mathrm{tr}(X^\dagger Y)$.

Proposition 1 (Split source at first order). *The J_Π-odd part of $E_\Pi^2|_{\mathbb{C}^3_{\rm gen}}$ at first order in s is the anticommutator $\{E_0, E_1\}$, so*

$$u(s) = s\,\frac{\langle J_3, \{E_0, E_1\}\rangle}{\langle J_3, J_3\rangle} + O(s^3), \qquad v(s) = s\,\frac{\langle R_{\rm mix}, \{E_0, E_1\}\rangle}{\langle R_{\rm mix}, R_{\rm mix}\rangle} + O(s^3). \tag{8}$$

For phase-free jet data, i.e. real-symmetric E_0, E_1, the anticommutator $\{E_0, E_1\}$ is symmetric, so $v \equiv 0$ identically while $\langle J_3, \{E_0, E_1\}\rangle$ is generically non-zero. A non-zero mixing coefficient v requires a complex metaplectic phase: a Hermitian J_Π-odd imaginary piece $\mathrm{i}\, a\, R_{\rm mix}$ in E_1.

Proof. Square the residue jet: $E_\Pi^2 = E_0^2 + s\,\{E_0, E_1\} + s^2(\{E_0, E_2\} + E_1^2) + O(s^3)$. By Lemma 1 the summands have J_Π-parity even, odd, even, so the odd part at order s is $\{E_0, E_1\}$. Projecting onto J_3 and $R_{\rm mix}$ gives (8). If E_0, E_1 are real symmetric then $\{E_0, E_1\}$ is real symmetric, hence Hilbert–Schmidt orthogonal to the antisymmetric $R_{\rm mix}$, so $v \equiv 0$; the pairing with the symmetric J_3 is unconstrained and generically non-zero. Adding $\mathrm{i}\, a\, R_{\rm mix}$ (Hermitian, J_Π-odd) to E_1 produces an a-dependent $\langle R_{\rm mix}, \cdot\rangle$, so v depends on the phase a. □

Proposition 1 reproduces, at the level of the constructed block, the two facts established upstream by other means: the vanishing of the mixing partner ($\mu \equiv 0$ in the metaplectic computation of [5]) and the statement that generation mixing requires the complex metaplectic phase [1].

6 Identification of the chiral generator

Theorem 1 and Proposition 1 fix the structural role of the generator A_- but leave its content implicit. We now identify it in the chiral frame of the Schur reduction [5]. In the chiral splitting $\mathcal{S}_\Pi = \mathcal{S}_L \oplus \mathcal{S}_R$ the Dirac operator is chirality-odd and the eliminated block has the block form [5]

$$D = \begin{pmatrix} 0 & \mathcal{D}^- \\ \mathcal{D}^+ & 0 \end{pmatrix}, \qquad 1 - P = \begin{pmatrix} \pi_{LL} & \pi_{LR} \\ \pi_{RL} & \pi_{RR} \end{pmatrix}, \tag{9}$$

with π_{LL}, π_{RR} self-adjoint and $\pi_{RL} = \pi_{LR}^\dagger$. The Born–Infeld parity J_Π is the antiunitary $J_\Pi = \epsilon \circ \mathrm{conj}$ that exchanges the chiralities, $J_\Pi P_L = P_R J_\Pi$, with unitary intertwiner $\tau : \mathcal{S}_R \to \mathcal{S}_L$ [5, 3].

Lemma 2 (Defect bridge, $\tau = \mathrm{id}$ frame). *In the frame $\tau = \mathrm{id}$, the left-handed diagonal block of the J_Π-odd part of the eliminated block is half the transported equivariance defect,*

$$[\Pi_{J_\Pi\text{-}odd}(1 - P)]_{LL} = \tfrac{1}{2}(\pi_{LL} - \overline{\pi_{RR}}) = \tfrac{1}{2}\Delta_\chi(P). \tag{10}$$

Proof. With $J_\Pi X J_\Pi^{-1} = \Sigma\,\overline{X}\,\Sigma$, where $\Sigma = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ is the chirality swap, a block computation gives $J_\Pi(1-P)J_\Pi^{-1} = \left(\begin{smallmatrix}\overline{\pi_{RR}}&\overline{\pi_{RL}}\\\overline{\pi_{LR}}&\overline{\pi_{LL}}\end{smallmatrix}\right)$, whose LL block is $\overline{\pi_{RR}}$. Subtracting from $1-P$ and halving gives (10). □

Proposition 2 (Defect bridge, general τ)**.** *For the general unitary intertwiner τ the identity holds in transported form,*

$$[\Pi_{J_\Pi\text{-}odd}(1-P)]_{LL} = \tfrac{1}{2}\big(\pi_{LL} - \tau\,\overline{\pi_{RR}}\,\tau^{-1}\big) = \tfrac{1}{2}\Delta_\chi(P), \tag{11}$$

obtained from Lemma 2 by unitary transport of the right-handed block by τ.

Remark 2. Lemma 2 is the case computed symbolically (Appendix A); Proposition 2 follows from it by the unitary conjugation τ and is stated as such, not as a separate computation.

Applying the bridge to the jet (3) order by order, the equivariant finite point has $\Delta_\chi(P(0)) = 0$, so $u(0) = 0$, while the first-order coefficient carries the rate

$$[\Pi_{J_\Pi\text{-odd}}\dot{Q}(0)]_{LL} = \tfrac{1}{2}\partial_s\Delta_\chi(P)\big|_0, \qquad \dot{Q}(0) = Q_1, \tag{12}$$

which by Theorem 1(ii) is the chiral-diagonal component of $[A_-, Q_0]$.

Corollary 1 (Identification of A_-)**.** *The chiral generator is the infinitesimal transported chiral equivariance defect of the eliminated block: the J_Π-odd tangent $[A_-, Q_0] = \Pi_{J_\Pi\text{-}odd}\dot{Q}(0)$ has chiral-diagonal component $\frac{1}{2}\partial_s\Delta_\chi(P)|_0$. Through Proposition 1 the split rate is*

$$u'(0) = \frac{\langle J_3, \{E_0, E_1\}\rangle}{\langle J_3, J_3\rangle}, \qquad E_1 = -\Pi_S\, D\,\dot{Q}(0)\,D\,\Pi_S^*, \tag{13}$$

so the carrier of the split is $\partial_s\Delta_\chi(P)|_0$, dressed by the $\mathcal{D}^\pm$ transport and projected to $\mathbb{C}^3_{\text{gen}}$ through the localisation $u \propto \pi_{RR} - \pi_{LL}$ of [5].

The generation projection to $\mathbb{C}^3_{\text{gen}}$ is not re-derived here: it is the localisation established in [5], used as an input. Two boundaries must be stated precisely. First, A_- is not an additional selector: its odd component is exactly the transported defect rate, an object already present in the reduction. Second, A4 is the only admissible stratum in which the infinitesimal defect may be non-zero [5]; the present identification fixes the defect rate as the unique structural carrier of the split, but does not determine its absolute value. In short, A_- is identified structurally, while $\partial_s\Delta_\chi(P)|_0$ is not normalised here.

7 Scope and consequence

Corollary 2 (Front C admissibility)**.** *The constrained jet admits $u \neq 0$ at order s with no R_{mix} leakage and with $|u|$ unconstrained. The split vanishes at the finite point, $u(0) = 0$, is carried solely by the chiral Grassmann tangent $[A_-, Q_0]$, and is invisible to the finite even block Q_0.*

The corollary is the precise sense in which the eliminated block "makes $u \neq 0$ possible without importing $|u|$". No object of the value dictionary — the chiral-frontier normalisation, the cascade exponent, the saturation selectors — appears anywhere in Theorem 1, Lemma 1, Proposition 1, or the generator identification of Section 6; the construction consumes only the Schur form (1), the projector constraints (2), the chiral block structure (9), the J_Π-grading, and the electric-genus sign $\mu_\chi^2 < 0$ [4] used solely to orient the deformation. With the generator identified as the transported defect rate (Corollary 1), the single remaining target is sharply specified: *determine the admissible normalisation of* $\partial_s \Delta_\chi(P)|_0$, the A4-level chiral defect rate that carries the *magnitude* of the split. This is the downstream, dictionary-bound question [1]; the present note fixes the *structure* of u and its unique carrier, not its value.

A Reproducibility

All identities are verified by exact computation, with no floating-point sampling, in `simulation/fermionic-matter/eliminated_block_jet.py`. The projector-jet identities of Theorem 1, including the opposite signs of the Q_2 diagonal blocks (7), are checked on independent generic rational instances of a J_Π-even projector Q_0 and an anti-self-adjoint generator A. The parity decomposition (6) of Theorem 1(ii) and Remark 1 are confirmed: a purely J_Π-even generator yields $Q_1 \neq 0$ with $\Pi_{J_\Pi\text{-odd}} Q_1 = 0$. The parity propagation of Lemma 1 is verified as a symbolic operator identity. On $\mathbb{C}^3_{\mathrm{gen}}$, the anticommutator $\{E_0, E_1\}$ is confirmed J_Π-odd, the mixing coefficient v vanishes identically for real-symmetric data, and a generic phase-free witness gives $u \neq 0$ with $v = 0$ and $|u|$ free. A sign correction surfaced during verification and is recorded in (7): the lower diagonal block of Q_2 carries $+(1 - Q_0)Q_1^2(1 - Q_0)$, opposite to the upper block $-Q_0 Q_1^2 Q_0$.

The generator identification of Section 6 is verified by exact complex-rational computation in `simulation/fermionic-matter/front_d0_generator.py`. The defect bridge (10) of Lemma 2 is checked in the $\tau = \mathrm{id}$ frame, together with the vanishing of the defect at the equivariant point, the chiral block transport $E_{LL} = -\Pi_S \mathcal{D}^- \pi_{RR} \mathcal{D}^+ \Pi_S^*$, $E_{RR} = -\Pi_S \mathcal{D}^+ \pi_{LL} \mathcal{D}^- \Pi_S^*$ of [5], and the rate identity (12) relating $\Pi_{J_\Pi\text{-odd}} \dot{Q}(0)$ to $\partial_s \Delta_\chi(P)|_0$. The general-$\tau$ Proposition 2 is obtained by unitary transport and is not separately computed.

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[5] Jérôme Beau. The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking, 2026. Fermionic matter sub-programme note: exhibits the projective endomorphism as the Schur complement $E_\Pi = -\Pi_S\, D\,(1 - P)\, D\, \Pi_S^* = -M^\dagger M$ of the spinorial directions eliminated by the non-injective projection; reduces the inter-generation split coefficient u to the $\mathcal{D}^\pm$-transported, generation-projected component of the antiunitary chiral equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau\, \overline{\pi_{RR}}\, \tau^{-1}$; stratifies the result on the Cosmochrony axioms (A1–A3 force $u = 0$; non-zero u requires A4 chiral symmetry-breaking); proves finite/Lorentzian separation (finite locking is J_Π-equivariant, so $u_{\mathrm{fin}} = 0$); locks Seeley–DeWitt conventions so operator and spectral u coincide; no-go result: chirality is Lorentzian, not a finite-fibre label. Closes the Schur transversality in the present Lorentzian spin stratum, so $u \neq 0$, doi:10.5281/zenodo.20601040.